Deep Learning Assignment - VGG6 on CIFAR-10

Q1: Baseline Training

Trained VGG6 on CIFAR-10 with data augmentation (random crop, horizontal flip) and normalization. Used Adam optimizer with learning rate 0.001, batch size 128, and 25 epochs.

Q2: Experimental Results

Config 1 (ReLU + Adam): 87.66%
Config 2 (Sigmoid + Adam): 10.00% (failed to converge)
Config 3 (Tanh + Adam): 86.06%
Config 4 (GELU + Adam): 90.98% (BEST)
Config 5 (ReLU + SGD, lr=0.01): 85.67%
Config 6 (ReLU + RMSprop): 85.01%
Config 7 (ReLU + Adam, batch=64): 86.99%
Config 8 (ReLU + Adam, lr=0.0001): 74.03%

Key Findings:
- GELU activation outperformed ReLU by 3.32%
- Sigmoid activation failed completely (vanishing gradients)
- Adam optimizer performed best overall
- Lower learning rate (0.0001) hurt performance
- Smaller batch size (64) slightly reduced accuracy

Q3: Visualizations

All plots generated using Weights & Biases (W&B) logging. Includes configuration comparison, learning rate analysis, and training curves for the best model.

Q4: Best Configuration

Activation: GELU
Optimizer: Adam
Learning Rate: 0.001
Batch Size: 128
Epochs: 25
Best Validation Accuracy: 90.98%

Q5: Code and Reproducibility

Complete code available in Google Colab notebook with W&B integration. All experiments logged with full hyperparameter tracking. Random seeds set for reproducibility.